

UST-ORDYNSKY, Russian Federation

WMO: 307130

Lat: 52.817N

Lon: 104.767E

Elev: 1724

StdP: 13.80

Time Zone: 8.00 (E08)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-27.9	-36.2	0.7	-32.5	-31.7	1.0	-27.7	20.7	-2.2	17.6	1.0	4.3	20	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	82.9	65.6	79.4	64.0	76.2	62.4	68.1	77.2	66.1	75.3	64.3	73.5	7.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	99.1	71.7	62.9	91.5	69.8	61.0	85.4	68.3	33.6	77.0	31.9	75.2	30.5	73.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.2	15.1	DB	-38.6	89.7	5.2	3.3	-42.3	92.1	-45.4	94.0	-48.3	95.8	-52.1	98.2
			WB	-38.3	71.6	5.0	2.6	-41.9	73.5	-44.9	75.1	-47.7	76.5	-51.3	78.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	-8.3	-3.7	14.0	32.6	47.5	58.1	63.3	59.6	47.1	32.7	10.8	-3.6
	DBStd	26.83	9.99	9.91	11.15	7.54	7.58	6.97	5.40	5.17	6.88	7.67	11.63	11.27
	HDD50	8615	1809	1505	1117	523	139	13	1	1	133	535	1177	1662
	HDD65	13090	2274	1925	1582	973	544	224	96	182	536	1000	1627	2127
	CDD50	1074	0	0	0	0	61	256	412	298	47	0	0	0
	CDD65	74	0	0	0	0	1	17	42	14	0	0	0	0
	CDH74	1146	0	0	0	0	67	371	506	186	16	0	0	0
	CDH80	243	0	0	0	0	7	109	94	33	0	0	0	0
Wind	WSAvg	6.4	5.5	5.5	6.3	8.6	8.4	7.0	5.8	5.5	6.2	6.1	5.9	5.6
Precipitation	PrecAvg	15.4	0.4	0.3	0.3	0.6	1.2	2.4	3.5	3.5	1.6	0.6	0.6	0.5
	PrecMax	23.5	0.9	0.6	0.9	1.8	2.2	6.0	7.9	7.6	4.3	1.4	1.2	1.2
	PrecMin	11.4	0.1	0.1	0.0	0.2	0.4	0.8	1.2	1.3	0.5	0.2	0.2	0.1
	PrecStd	2.8	0.2	0.1	0.2	0.5	0.5	1.2	1.4	1.7	0.9	0.3	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	28.4	52.9	63.0	81.2	88.6	85.9	84.2	76.9	61.3	41.0	25.9
		MCWB	16.8	25.1	41.5	47.6	59.7	65.0	67.7	67.7	61.3	48.7	35.2	24.5
	2%	DB	13.5	19.4	39.9	56.6	75.9	83.1	82.7	79.1	69.8	55.1	34.2	19.9
		MCWB	12.6	18.1	33.8	43.5	56.1	64.0	66.8	65.5	57.4	45.3	30.6	18.8
	5%	DB	9.3	15.2	35.3	51.6	71.0	78.0	79.6	75.3	64.6	49.8	30.0	16.0
		MCWB	8.5	13.9	31.1	40.3	53.6	61.7	65.4	63.5	53.6	42.6	27.5	15.1
	10%	DB	5.8	11.3	31.3	46.9	66.1	73.9	76.5	71.8	60.2	45.0	26.9	12.8
		MCWB	5.1	10.2	28.0	37.5	50.9	59.4	64.4	61.7	51.5	39.6	25.2	11.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	25.3	42.0	48.3	60.9	69.1	71.2	70.3	61.4	49.6	35.5	25.1
		MCDB	18.1	27.8	53.2	60.1	79.5	79.6	80.9	80.9	73.5	59.4	39.5	26.6
	2%	WB	12.3	17.9	34.5	44.4	57.4	65.7	68.9	66.9	58.0	46.2	31.2	18.7
		MCDB	13.4	19.6	39.0	56.2	73.6	77.3	78.1	75.9	67.4	54.2	33.9	19.6
	5%	WB	8.5	14.0	31.3	41.0	54.6	63.5	67.0	64.6	55.1	43.1	27.8	15.2
		MCDB	9.2	15.3	35.3	50.9	69.2	75.3	75.8	73.7	62.8	48.8	29.8	15.9
	10%	WB	5.1	10.3	28.1	38.2	51.2	61.4	65.3	62.6	52.5	40.2	25.0	12.0
		MCDB	5.7	11.5	31.4	46.2	64.6	71.7	73.9	69.9	58.5	44.8	26.9	12.7
Mean Daily Temperature Range	5% DB	MDBR	15.5	19.8	23.4	21.1	25.5	22.4	20.1	19.4	20.7	20.6	18.2	15.6
		MCDBR	17.7	21.1	23.2	30.1	36.4	33.4	27.8	27.4	30.0	28.7	21.0	17.1
		MCWBR	16.9	19.8	17.1	18.7	19.8	16.2	13.7	15.0	16.9	18.8	17.7	16.3
	5% WB	MCDBR	17.8	21.2	22.6	28.1	34.1	25.7	23.9	24.9	26.5	27.2	20.4	17.3
		MCWBR	16.9	19.8	17.0	18.1	18.8	14.0	13.3	14.0	15.7	18.5	17.5	16.6
Clear-Sky Solar Irradiance	taub	0.272	0.297	0.329	0.375	0.432	0.439	0.427	0.387	0.338	0.315	0.294	0.274	
	taud	2.083	2.070	2.103	2.223	2.086	2.171	2.248	2.345	2.438	2.400	2.211	2.072	
	Ebn at Noon	218	251	268	271	259	259	260	265	267	248	210	191	
	Edh at Noon	23	32	39	40	48	44	41	35	28	23	21	20	
All-Sky Solar Radiation	RadAvg	395	756	1248	1570	1760	1866	1721	1478	1090	690	389	274	
	RadStd	20	25	49	95	107	103	136	64	99	41	23	21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page